

Influence Maximization When Communication Is the Bottleneck

From centralized submodular greedy to CELF acceleration and local candidate sharing

Yixin Chen Bichi Zhang Yaoqin Ye

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Research question: how much influence quality can we keep when expensive centralized seed selection is replaced by cached greedy evaluation or communication-limited local candidates?

1. Optimization Target

We choose k initial seeds in a network and maximize expected diffusion coverage:

$$\max_{S \subseteq V, |S| \leq k} \sigma(S).$$

Here $\sigma(S)$ is estimated by Monte Carlo simulation under the Independent Cascade model.

2. Diffusion Model

- ▶ Newly activated nodes get one chance to activate each inactive out-neighbor.
- ▶ Each edge attempt succeeds independently with probability p_{uv} .
- ▶ We use the weighted-cascade rule:

$$p_{uv} = \frac{1}{\text{indegree}(v)}.$$

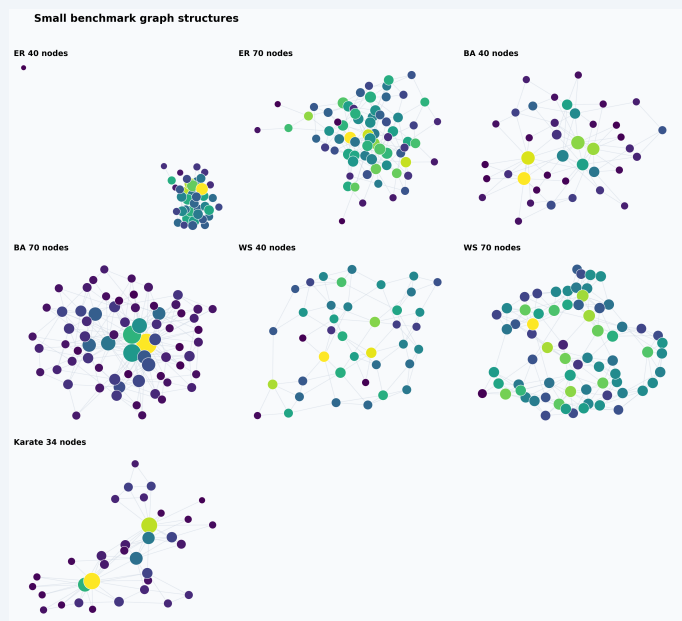
3. Why Greedy Is Principled

IC influence spread is monotone and submodular:

$$\sigma(A \cup \{v\}) - \sigma(A) \geq \sigma(B \cup \{v\}) - \sigma(B).$$

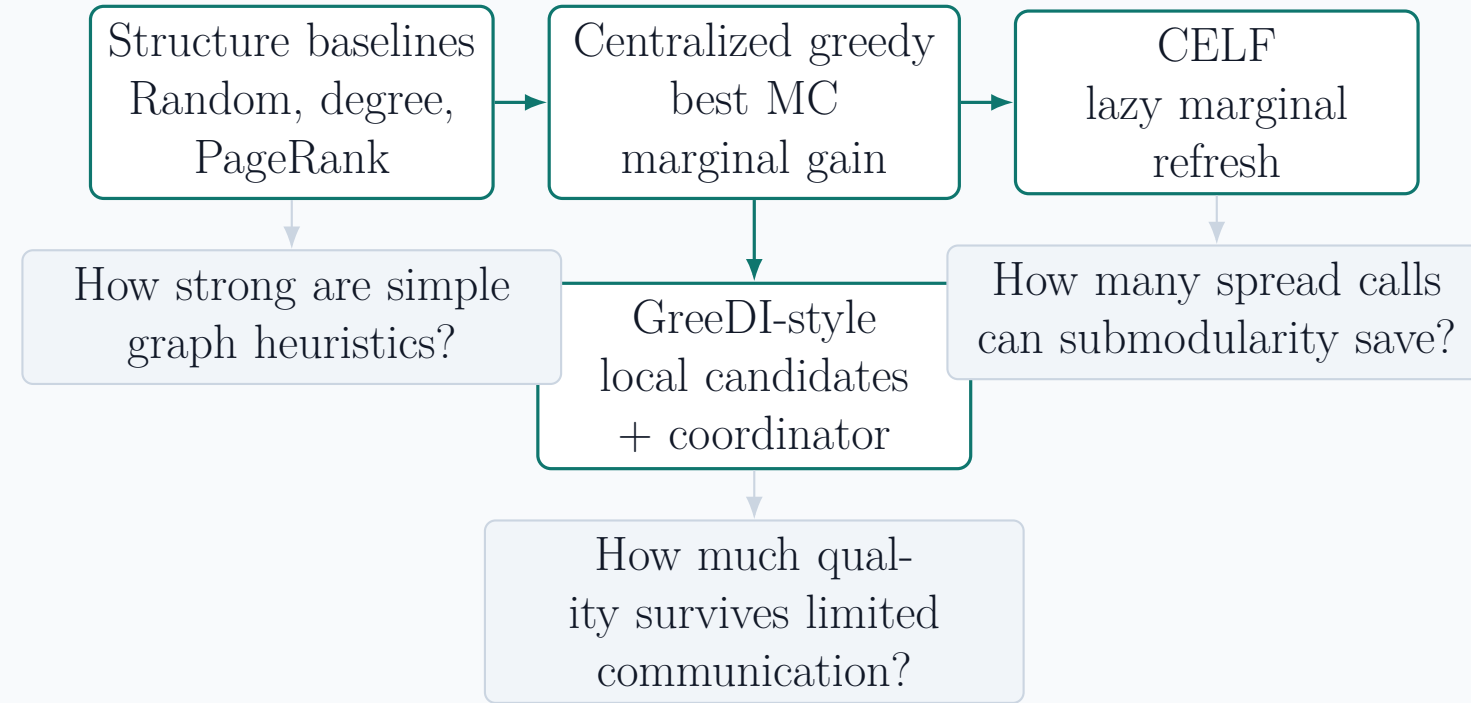
- ▶ Exact influence maximization is NP-hard.
- ▶ Greedy has a $(1 - 1/e)$ guarantee for the exact objective.
- ▶ The bottleneck is not the rule itself; it is repeated spread estimation.

Benchmark Graphs

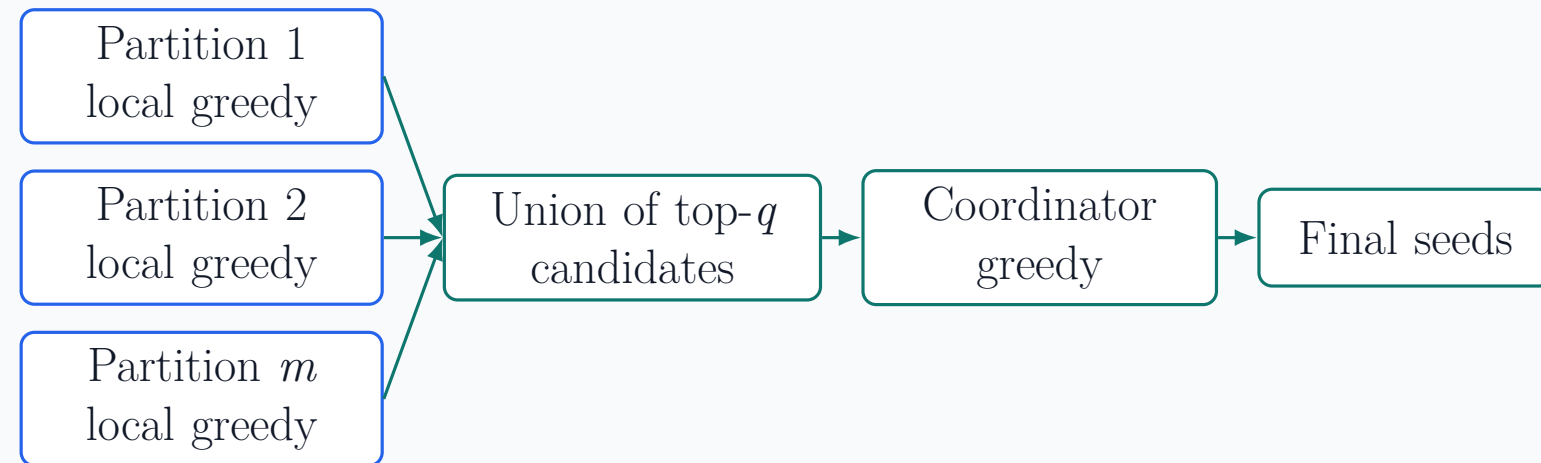


Random, hub-heavy, small-world, and Karate Club graph structures expose different seed-selection behavior.

Algorithmic Story



Distributed Candidate Sharing



Each worker sends at most q candidate nodes. The coordinator searches only the merged candidate pool.

Experimental Protocol

Graph families	Erdoes-Renyi, Barabasi-Albert, Watts-Strogatz, Karate
Large sizes	$n \in \{50, 100, 200\}$, plus Karate
Seed budget	$k = 10$ for synthetic large graphs, $k = 5$ for Karate
Distributed sweep	$m \in \{2, 4, 8\}$, $q \in \{k, 2k, 5k\}$
Fair evaluation	final seed sets re-estimated with a common seed

74.1% fewer evaluations

CELFF vs. greedy

Large-scale average: 266.2 vs. 1026.5 spread evaluations.

83.5% lower runtime

CELFF vs. greedy

Large-scale average: 0.77s vs. 4.67s wall-clock time.

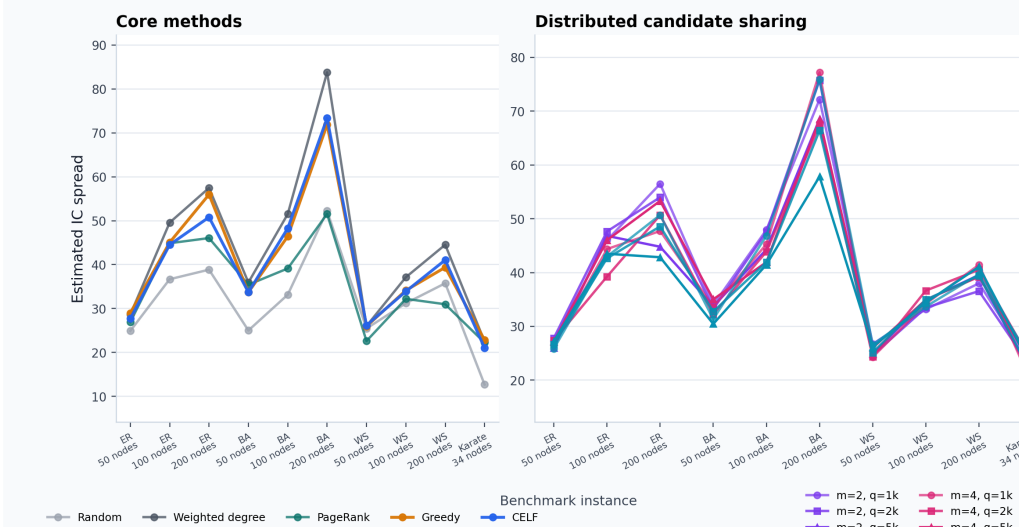
Large-Scale Evidence

Method	Spread	Ratio	Time	Eval.
Random	31.59	0.791	0.00s	1.0
W. Degree	43.71	1.066	0.01s	1.0
PageRank	35.19	0.893	0.00s	1.0
Greedy	40.43	1.000	4.67s	1026.5
CELFF	40.05	0.988	0.77s	266.2

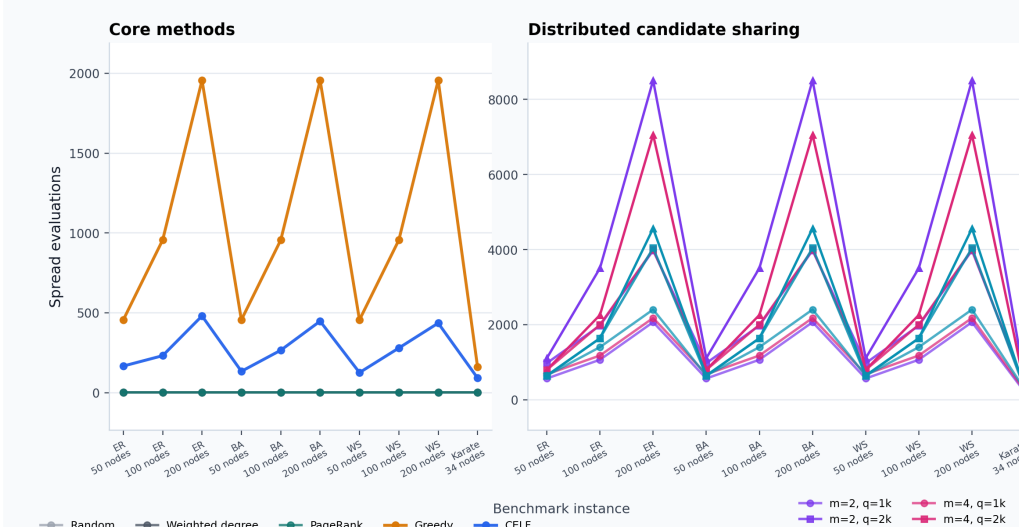
- ▶ CELFF nearly preserves greedy spread but sharply cuts spread calls.
- ▶ Weighted degree is not a weak baseline under weighted-cascade probabilities.
- ▶ Distributed quality remains competitive, but communication cost rises quickly.

Case-Level Results

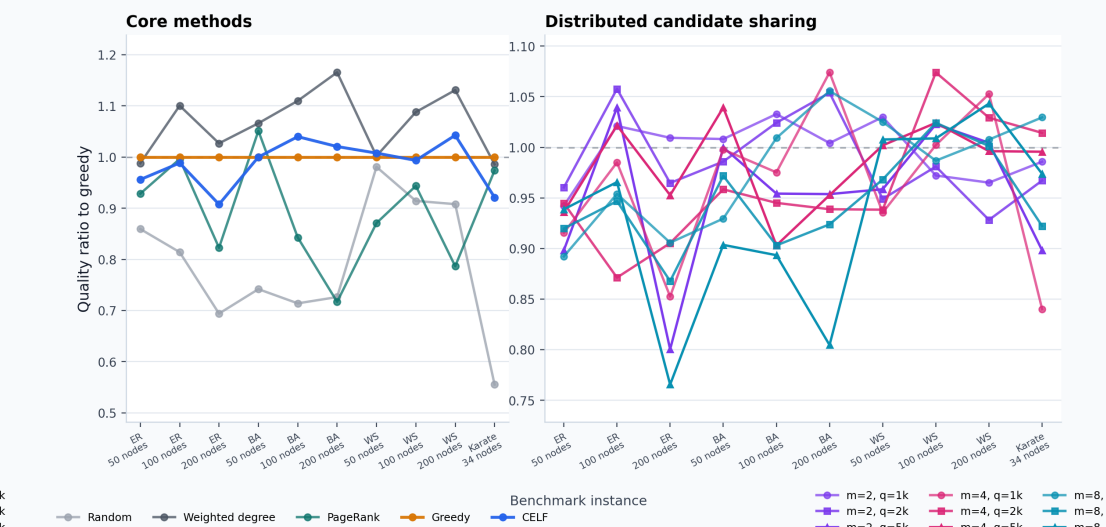
Estimated IC spread



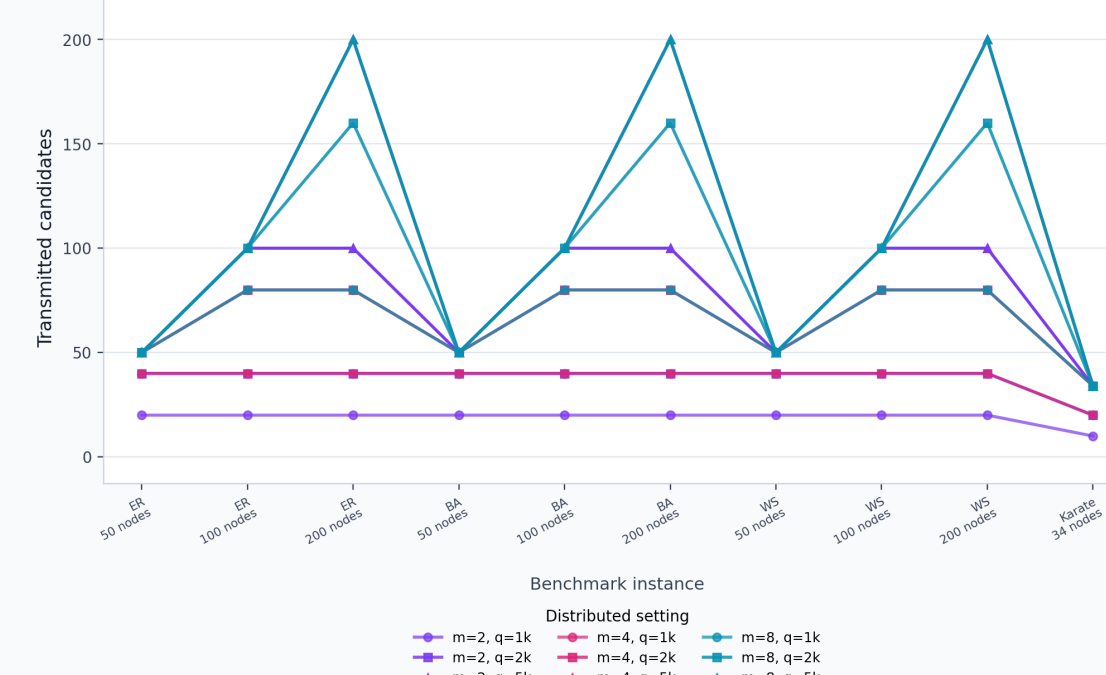
Spread evaluations



Quality ratio to greedy



Transmitted candidates



Communication Tradeoff

Budget	Ratio	Time	Eval.	Cand.
$q = k$	0.980	1.58s	1242.4	41.1
$q = 2k$	0.965	3.36s	2037.2	66.9
$q = 5k$	0.955	6.48s	3044.4	98.4

Bigger candidate budgets transmit more nodes and trigger more evaluations. In this noisy course-scale setting, quality does not improve monotonically with q .

Takeaways

- ▶ Submodularity gives the project a principled greedy baseline.
- ▶ CELFF is the clearest algorithmic improvement: it keeps the target but removes redundant evaluations.
- ▶ Communication-limited candidate sharing changes the cost model: less global search, more sensitivity to partitions and Monte Carlo noise.

Implementation. IC simulation, greedy selection, CELFF, and distributed candidate sharing are implemented as separate modules.

Run command. `python src/experiments.py --scale large --seed 7 --simulations 40`

Scope. This is a course-scale reproduction of algorithmic mechanisms, not a full IMM/RIS or production distributed system.